Ben Lonnqvist

Rue Saint-Laurent 33-35, 1003 Lausanne, Switzerland
ben.lonnqvist@epfl.ch | Google Scholar

Education

2020- Ph.D. student, Neuroscience
EPFL (École polytechnique fédérale de Lausanne) | Lausanne, Switzerland
- Laboratory of Psychophysics–PI: Prof Michael Herzog
- Primary interests: neural network modelling of vision; visual crowding; experimental design
- Teaching: Analysis I Fall 2020 teaching assistant

2016-2020 M.A. Hons, Economics and Finance
University of Aberdeen | Aberdeen, Scotland
- First-class Honours degree
- Quantitative focus in electives
- Relevant coursework: Econometrics, Mathematical and Statistical Methods in Economics, Stochastic processes, Understanding Statistics, Proof-based Microeconomics

2011-2014 Vocational Qualification in Business Administration
Omnia Vocational College | Espoo, Finland
- Student coach for Skills Entrepreneurship, 2013

Research Work Experience

Mar - May 2018 - *2020* Aberdeen School of Psychology | Undergraduate Research Assistant
- Voluntary research assistant in the Consciousness, Attention, and Perception Lab
- First-authored a paper that has now been published in *Neural Networks*
- Using Python, Keras and NumPy, programmed several architectures of Deep Neural Networks and wrote testing environments to produce novel research on visual crowding in convolutional NNs
- Used SSH tunneling to access a Linux high-power computing cluster

May - Aug 2019 - *2019* Aberdeen School of Psychology | Research Intern
- Summer research intern at the Eye Movements and Attention Lab
- Pre-print under review by *PLOS Computational Biology*: https://psyarxiv.com/suj28/
- Using Python, modeled Bayes-optimal eye movements parameterized by human visual search data
- Using MATLAB and PsychToolbox, wrote the majority of a visual search psychophysical experiment
- Ran 65 participant-hours of eye tracking experiments using EyeLink hardware

Research

Publications

2020 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Crowding in humans is unlike that in convolutional neural networks. Publisher: *Neural Networks*. DOI: https://doi.org/10.1016/j.neunet.2020.03.021

Pre-prints

2020 **Lonnqvist, B.**, Elsner, M., Hunt, A. R., Clarke, A. D. F. (2020). Modeling individual variation in visual search with reinforcement learning. PsyArXiv. DOI: https://doi.org/10.31234/osf.io/suj28. Submitted to *PLOS Computational Biology*.

Conference abstracts

2019 **Lonnqvist, B.**, Clarke, A. D. F., Chakravarthi, R. (2019). Object Recognition in Deep Convolutional Neural Networks is Fundamentally Different to That in Humans. Abstract presented at the 18th annual meeting of the Scottish Vision Group, Scotland, UK.

Invited talks

2019 **Lonnqvist, B.**, Clarke, A. D. F., Hunt, A. (2019). Changes in Human Eye Movement Strategies Over Time. Talk presented at a meeting of the schools of Psychology and Computer Science at the University of Essex, Britain, UK.

Awards

Rank Prize Foundation Optoelectronics Studentship

2019 A £3000 studentship awarded for a summer research project.

Research visit support by the James S McDonnell Foundation Scholar Award in Human Cognition (PI: Dr A. Hunt)

2019 A £150 support for a research visit to the University of Essex in July 2019.

Skills Finland Entrepreneurship National Finalist

2012 Competed in the Finnish National Finals of the Skills competition in 2012.

Technical

Mathematics:	Optimisation, Multiv. Calculus, Linear Algebra, Bayesian Statistics
Strong:	python, SciPy ecosystem, Pandas, Statsmodels, Keras, LaTeX
Familiar:	matlab, C#, Unity, TensorFlow, AWS

References

Michael H. Herzog, Full Professor
EPFL (École polytechnique fédérale de Lausanne) | Lausanne, Switzerland
PI: Laboratory of Psychophysics, Brain Mind Institute
e-mail: michael.herzog@epfl.ch

Amelia Hunt, Senior Lecturer (Associate Professor)
University of Aberdeen | Aberdeen, Scotland
PI: Eye Movements and Attention Lab, Department of Psychology
e-mail: a.hunt@abdn.ac.uk

Ramakrishna Chakravarthi, Lecturer (Assistant Professor)
University of Aberdeen | Aberdeen, Scotland
PI: Consciousness, Attention, and Perception Lab, Department of Psychology
e-mail: rama@abdn.ac.uk